

Linear Algebra (MTS203)

Final Exam, (180 mins — 60 Marks)

Class no. 7500 (Fall 2023)

Exam Date - Dec 23, 2023



Institute of
Business Administration
Karachi

Leadership and Ideas for Tomorrow

COMPLETE THE FOLLOWING CAREFULLY AND CLEARLY:

(Please Print)

Full Name: _____

E.R.P. Number: _____ Invigilator's Signature: _____

This exam has EIGHTEEN pages, excluding this one. Questions (8 pages), solution work (10 empty pages).

The questions must be answered in the spaces provided. Show details of all solutions. Writing must be clear and legible. If response to a question overflows on to subsequent sheets, it must be clearly indicated in the main response space.

Under no circumstances, may a student take this exam out of the exam room.

On no account are pages to be torn or removed from the exam, unless specifically authorized by the invigilator.

Detection of usage of any unfair means will result in immediate disqualification as per IBA policy.

Candidates must not have in their possession, books, notes or papers of any kind, nor any portable electronic devices, except calculators.

MARKS

Question 1. _____ /12

Question 2. _____ /12

Question 3. _____ /12

Question 4. _____ /12

Question 5. _____ /12

Question 6. _____ /12

TOTAL _____ /60

FOR INSTRUCTOR'S USE ONLY

FINAL ____%	MIDTERM ____%	other work ____%	TOTAL 100	Instructor's Signature

Question - 1a (5 marks)

In the box preceding each of the following statements, write T for true or F for false, as your true/false response to the given statement.

- (a) ☐ A full rank $n \times m$ matrix, with $n > m$, may have a right inverse, even if its transpose doesn't have a left inverse.
- (b) ☐ An over-determined linear system may have a unique solution.
- (c) ☐ The *row-rank* of a matrix, may be different from its *column-rank*.
- (d) ☐ An over-determined linear system's coefficient matrix may be full-rank.
- (e) ☐ An under-determined linear system may have a unique solution.
- (f) ☐ If A is singular and B is not, then AB must be singular.
- (g) ☐ There are infinitely many square matrices that are their own inverses.
- (h) ☐ If A and B are non-singular $n \times n$ matrices, then $A + B$ is also non-singular.
- (i) ☐ Elementary row operations can change the shape of a matrix.
- (j) ☐ A singular matrix A can be decomposed into a product of elementary matrices in at least two different ways, i.e., at least two matrices in the decomposition are different.

Question - 1b (4 marks)

Given

$$A = \begin{bmatrix} \mathcal{I} & 0 & 0 \\ B & \mathcal{I} & 0 \\ 0 & 0 & \mathcal{I} \end{bmatrix}$$

where all of the sub-matrices are $n \times n$, determine the block form of A^{-1} .

Solution:

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Question - 1c (3 marks)

Let $X = [2 \ 1 \ 5]^T$, and let $Y = [1 \ 2 \ 4]^T$. Calculate both outer products between them.

Solution:

Question - 2a (5 marks)

In the box preceding each of the following statements, write T for true or F for false, as your true/false response to the given statement.

- (a) ☐ Multiplying a singular matrix M to the left of both sides of an inconsistent system $Au = b$ will make it consistent.
- (b) ☐ $\det(AB) = \det(BA)$
- (c) ☐ $\det(A - B) = 0 \Rightarrow A = B$
- (d) ☐ $\det((AB)^T) = \det(A) \det(B)$
- (e) ☐ A square matrix in triangular form is nonsingular if and only if its diagonal entries are all nonzero.
- (f) ☐ If A is a nonsingular $n \times n$ matrix, then AA^T is also nonsingular.
- (g) ☐ If A and B are row equivalent matrices, then their determinants are equal.
- (h) ☐ If $\mathbf{x}$ is a nonzero vector in $\mathbb{R}^n$ and $A\mathbf{x} = \mathbf{0}$, then $\det(A) = 0$.
- (i) ☐ It is possible for a matrix to be symmetric, as well as skew-symmetric simultaneously.
- (j) ☐ Calculating the determinant of a matrix through row operations, involves lesser amount of work than calculating it through the so-called adjoint method.

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Question - 2b (4 marks)

Let A be an $n \times n$ matrix. Is it possible for $A^2 + I = 0$ in the case where n is odd? Answer the same question in the case where n is even.

Solution:

Question - 2c (3 marks)

Let $\mathbf{x}$ and $\mathbf{y}$ be distinct vectors in $\mathbb{R}^n$ (i.e., $\mathbf{x} \neq \mathbf{y}$), and let A be an $n \times n$ matrix with the property that $A\mathbf{x} = A\mathbf{y}$. Show that $\det(A) = 0$.

Solution:

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Question - 3a (5 marks)

In the box preceding each of the following statements, write T for true or F for false, as your true/false response to the given statement.

- (a) ☐ If U is the reduced row echelon form of A , then A and U have the same row space.
- (b) ☐ If $f_1, f_2, \dots, f_n$ are linearly dependent, then their Wronskian, i.e., $W[f_1, f_2, \dots, f_n]$ must be 0.
- (c) ☐ The set of polynomials in P_4 of odd degree, is a subspace of P_4 .
- (d) ☐ $C[a, b]$ is a subspace of $C^1[a, b]$.
- (e) ☐ If a given linear combination of n vectors, i.e., $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$ and $c_1 = c_2 = \dots = c_n = 0$ for $i = 1, 2, 3, \dots, n$, then these n vectors are linearly independent.
- (f) ☐ A finite dimensional vector space is one that is spanned by a finite basis.
- (g) ☐ If $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ span $\mathbb{R}^n$, with $m \neq n$, then they are linearly dependent.
- (h) ☐ Every basis is a spanning set, and, every spanning set is a basis.
- (i) ☐ If A and B are square matrices of the same order and rank, then the rank of A^2 and B^2 is the same.
- (j) ☐ If A is a fixed vector in $\mathbb{R}^{n \times n}$, then the set of all $n \times n$ matrices that commute with A forms a subspace of $\mathbb{R}^{n \times n}$.

Question - 3b (4 marks)

Let A be a 6×4 matrix of rank 4.

- a) What is the dimension of $\mathcal{N}(A)$? What is the dimension of the column space $\mathcal{R}(A)$?
- b) Do the column vectors of A span $\mathbb{R}^6$? Are the column vectors of A linearly independent? Write a (no more than) two sentence explanation for it.
- c) How many solutions will the linear system $A\mathbf{x} = \mathbf{b}$ have if $\mathbf{b}$ is in the column space of A ?

Solution:

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Question - 3c (3 marks)

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ be linearly independent vectors in $\mathbb{R}^n$, and let A be a nonsingular $n \times n$ matrix. Define $\mathbf{y}_i = A\mathbf{x}_i$ for $i = 1, \dots, k$. Show that $\mathbf{y}_1, \dots, \mathbf{y}_k$ are linearly independent.

Solution:

Question - 4a (5 marks)

In the box preceding each of the following statements, write T for true or F for false, as your true/false response to the given statement.

- (a) ☐ Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear operator. If $L(\mathbf{x}_1) = L(\mathbf{x}_2)$, then the vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ must be equal.
- (b) ☐ If L_1 and L_2 are linear operators on a vector space V , then $L_1 + L_2$ is also a linear operator on V , where $L_1 + L_2$ is the mapping defined by $(L_1 + L_2)(\mathbf{v}) = L_1(\mathbf{v}) + L_2(\mathbf{v})$ for all $\mathbf{v} \in V$.
- (c) ☐ If $L : V \rightarrow V$ is a linear operator and $\mathbf{x} \in \ker(L)$, then $L(\mathbf{v} + \mathbf{x}) = L(\mathbf{v})$ for all $\mathbf{v} \in V$.
- (d) ☐ If L_1 rotates each vector $\mathbf{x} \in \mathbb{R}^2$ by 80° and then reflects the resulting vector about the

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x -axis, and if L_2 is a transformation that does the same two operations but in the reverse order, then $L_1 = L_2$.

- (e) ☐ Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear operator, and let A be the standard matrix representation of L . If L^2 is defined by: $L^2(\mathbf{x}) = L(L(\mathbf{x}))$ for all $\mathbf{x} \in \mathbb{R}^2$, then L^2 is a linear operator and its standard matrix representation is A^2 .
- (f) ☐ The kernel of a linear transformation and the nullspace of the matrix representing the linear transformation are the same spaces.
- (g) ☐ The range of a linear transformation and the row-space of the matrix representing it are the same spaces.
- (h) ☐ If A is an $n \times m$ matrix, and then the linear transformation L_A representing the matrix maps $\mathbb{R}^n$ into $\mathbb{R}^m$.
- (i) ☐ Let $E = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ be an ordered basis for $\mathbb{R}^n$. If $L_1 : \mathbb{R}^n \rightarrow \mathbb{R}^n$, and $L_2 : \mathbb{R}^n \rightarrow \mathbb{R}^n$ have the same matrix representation with respect to E , then $L_1 = L_2$.
- (j) ☐ The matrix representation of a linear transformation is not unique.

Question - 4b (7 marks)

Let L be the linear operator on $\mathbb{R}^3$ defined by

$$L(\mathbf{x}) = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \end{bmatrix}$$

and let $S = \text{Span}(1, 0, 1)^T$. Find the kernel of L , and determine $L(S)$.

Solution:

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Question - 5a (5 marks)

In the box preceding each of the following statements, write T for true or F for false, as your true/false response to the given statement.

- (a) ☐ If $\mathbf{x}$ and $\mathbf{y}$ are nonzero vectors in $\mathbb{R}^n$, then the vector projection of $\mathbf{x}$ onto $\mathbf{y}$ is equal to the vector projection of $\mathbf{y}$ onto $\mathbf{x}$.
- (b) ☐ If $\mathbf{x}$ and $\mathbf{y}$ are unit vectors in $\mathbb{R}^n$ and $|\mathbf{x}^T \mathbf{y}| = 1$, then $\mathbf{x}$ and $\mathbf{y}$ are linearly independent.
- (c) ☐ If U , V , and W , are subspaces of $\mathbb{R}^3$, and if $U \perp V$, and $V \perp W$, then $U \perp W$.
- (d) ☐ It is possible to find a nonzero vector $\mathbf{y}$ in the column space of A such that $A^T \mathbf{y} = \mathbf{0}$.
- (e) ☐ If A is an $m \times n$ matrix, then AA^T and $A^T A$ have the same rank.
- (f) ☐ If an $m \times n$ matrix A has linearly dependent columns and $\mathbf{b}$ is a vector in $\mathbb{R}^m$, then $\mathbf{b}$ does not have a unique projection onto the column space of A .
- (g) ☐ If $\mathcal{N}(A) = \{\mathbf{0}\}$, then the system $A\mathbf{x} = \mathbf{b}$ will have a unique least squares solution.
- (h) ☐ The row space of A^T and the column space of A are the exact same vector spaces.
- (i) ☐ The nullspace of a matrix, is the orthogonal complement of the row space of A .
- (j) ☐ $\mathcal{R}(A^T)$ is the synonym of column space of A^T .

Question - 5b (7 marks)

For $A\mathbf{x} = \mathbf{b}$, find all least squares solutions, and show that each of these $\hat{\mathbf{x}}$ may be written as a sum of two vectors $\mathbf{y}$ and $\mathbf{z}$ such that $\mathbf{y} \in \mathcal{R}((A^T A)^T)$ and $\mathbf{z} \in \mathcal{N}(A^T A)$.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}, \text{ and } \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Solution:

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Question - 6a (5 marks)

In the box preceding each of the following statements, write T for true or F for false, as your true/false response to the given statement.

- (a) ☐ If A is an $n \times n$ matrix whose eigenvalues are all nonzero, then A is nonsingular.
- (b) ☐ If A is an $n \times n$ matrix, then A , and A^T have the same eigenvectors.
- (c) ☐ If A and B are similar matrices, then they have the same eigenvalues.
- (d) ☐ *Algebraic multiplicity* of an eigenvalue is the number of eigenvectors that belong to it.
- (e) ☐ *Geometric multiplicity* of an eigenvalue is the number of times it gets repeated as a root of the characteristic polynomial.
- (f) ☐ If A has eigenvalues of algebraic multiplicity greater than 1, then A must be singular.
- (g) ☐ If A is a 4×4 matrix of rank 3, and $\lambda = 0$ is an eigenvalue of algebraic multiplicity 3, then A is singular.
- (h) ☐ A singular matrix A of order $n \times n$ may have n linearly independent eigenvectors.
- (i) ☐ The product of all eigenvalues of a square matrix is called its trace.
- (j) ☐ The product of all eigenvalues of a square matrix is called its determinant.

Question - 6b (7 marks)

Let

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 2 & -2 \end{bmatrix}$$

Find all eigenvalues of A . For each eigenvalue, find a basis for the corresponding eigen-space.

Solution:

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